

✓ **Projet Epidémiologie Cabanes / Dechenais**

✓ **Chargement des données :**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.utils import resample
from imblearn.over_sampling import SMOTE
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from scipy import stats
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.impute import KNNImputer

%matplotlib inline

df_h1 = pd.read_csv('liver.csv')
df_h2 = pd.read_json('liver.json')
```

df_h1

	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	ALT	AST	Total_Proteins	Albumin	Albumin_and_Globulin_f
0	41 to 60 yo	Male	7.0	2.0	174.0	NaN	14	7800.0 mg/dL	4.2	
1	41 to 60 yo	Male	6.0	2.0	245.0	22.0	24	7100.0 mg/dL	3.4	
2	21 to 40 yo	Male	11.0	5.0	191.0	37.0	41	7700.0 mg/dL	4.3	
3	21 to 40 yo	Male	24.0	10.0	340.0	25.0	21	8300.0 mg/dL	4.5	
4	21 to 40 yo	Male	6.0	2.0	202.0	NaN	41	8000.0 mg/dL	3.9	
...	...	...	...	...	...	...	...	...	...	
316	<20 yo	Male	14.0	5.0	269.0	58.0	45	6700.0 mg/dL	3.9	
317	61 to 80 yo	Male	7.0	1.0	196.0	20.0	35	5800.0 mg/dL	2.0	
318	61 to 80 yo	Male	28.0	13.0	250.0	NaN	29	2700.0 mg/dL	0.9	
319	41 to 60 yo	Male	9.0	3.0	901.0	23.0	17	6200.0 mg/dL	3.5	
320	21 to 40 yo	Male	12.0	3.0	498.0	28.0	25	7000.0 mg/dL	3.0	

321 rows × 11 columns

Next steps:

[Generate code with df_h1](#)

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df_h2

	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Total
0	61 to 80 yo	Female	0.7	0.1	187.0	NaN	18	
1	61 to 80 yo	Male	10.9	5.5	699.0	64.0	100	
2	61 to 80 yo	Male	7.3	4.1	490.0	NaN	68	
3	41 to 60 yo	Male	1.0	0.4	182.0	14.0	20	
4	61 to 80 yo	Male	3.9	2.0	195.0	27.0	59	
...	...	...	...	...	...	...	...	
286	21 to 40 yo	Female	0.8	0.2	192.0	15.0	12	
287	41 to 60 yo	Male	1.3	0.6	155.0	15.0	20	
288	41 to 60 yo	Female	1.0	0.5	239.0	16.0	39	
289	41 to 60 yo	Male	4.5	2.3	315.0	120.0	105	
290	41 to 60 yo	Female	1.0	0.3	250.0	48.0	44	

291 rows × 11 columns

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▼ Analyse des jeux de données :

```
print(df_h1.info())
df_h2.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 321 entries, 0 to 320
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   Age                                  321 non-null   object
1   Gender                              321 non-null   object
2   Total_Bilirubin                     317 non-null   float64
3   Direct_Bilirubin                    320 non-null   float64
4   Alkaline_Phosphotase                317 non-null   float64
5   ALT                                 254 non-null   float64
6   AST                                 321 non-null   int64
7   Total_Proteins                      321 non-null   object
8   Albumin                             316 non-null   float64
9   Albumin_and_Globulin_Ratio          262 non-null   float64
10  Result                              321 non-null   int64
dtypes: float64(6), int64(2), object(3)
memory usage: 27.7+ KB
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 291 entries, 0 to 290
```

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
0	Age	291 non-null	object
1	Gender	291 non-null	object
2	Total_Bilirubin	290 non-null	float64
3	Direct_Bilirubin	287 non-null	float64
4	Alkaline_Phosphotase	290 non-null	float64
5	Alamine_Aminotransferase	236 non-null	float64
6	Aspartate_Aminotransferase	291 non-null	int64
7	Total_Proteins	289 non-null	float64
8	Albumin	290 non-null	float64
9	Albumin_and_Globulin_Ratio	227 non-null	float64
10	Outcome	291 non-null	int64

dtypes: float64(7), int64(2), object(2)

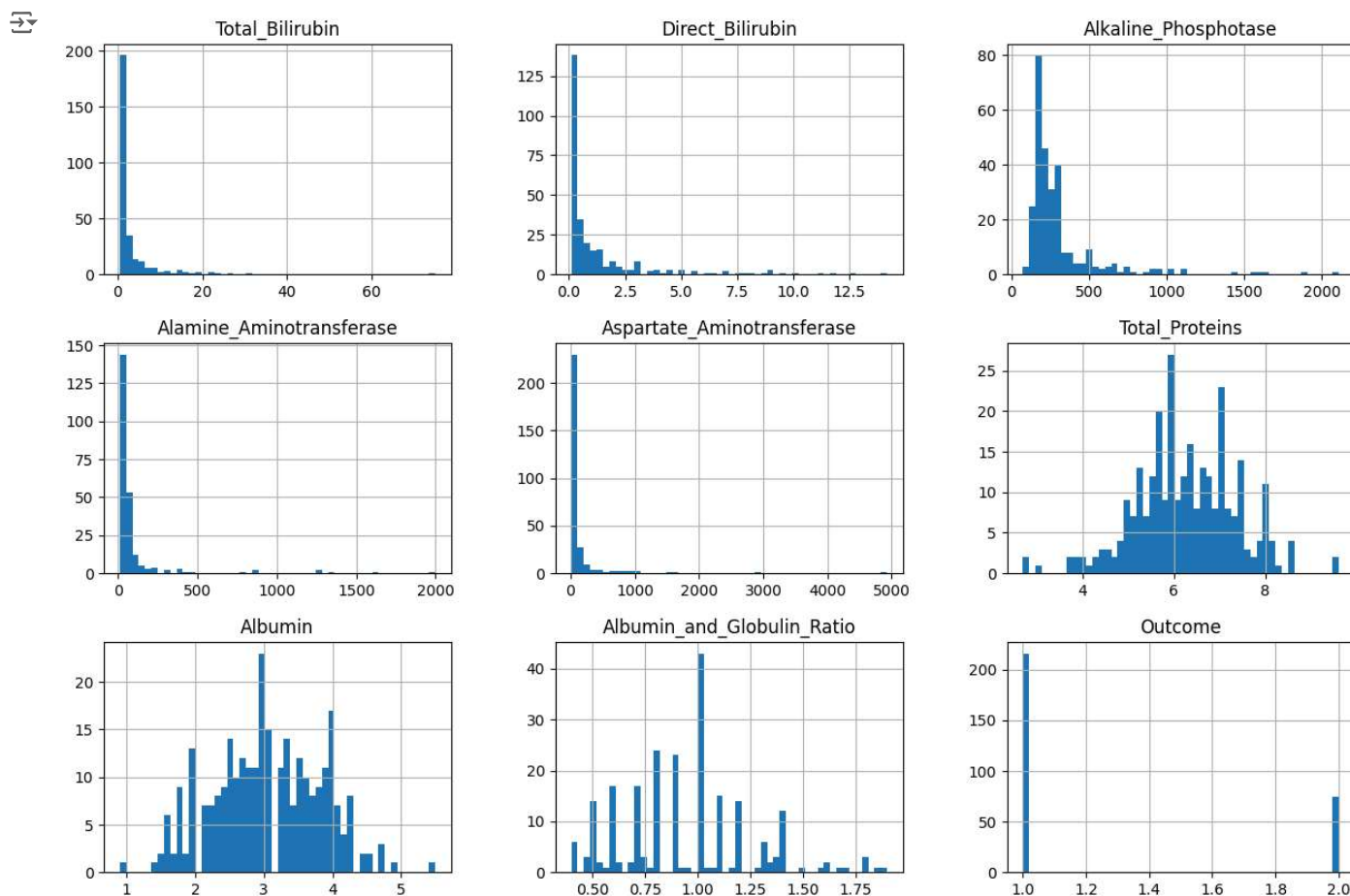
memory usage: 25.1+ KB

Remarque :

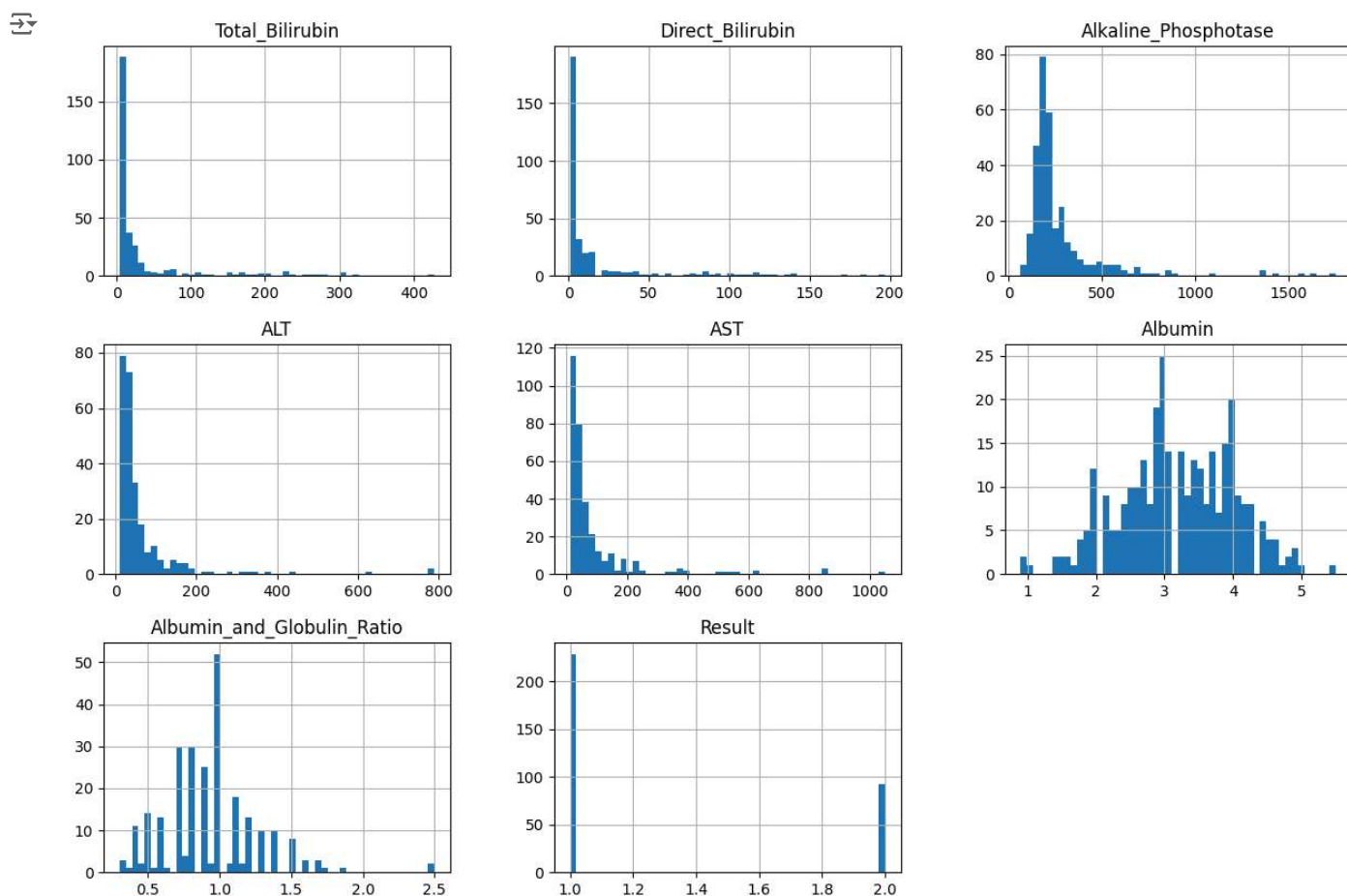
- Total_Proteins est de type objet dans le df_h1 et de type float64 dans le df_h2
- ALT, ASP et Result de df_h1 sont appelés Alamine_Aminotransferase, Aspartate_Aminotransferase et Outcome dans df_h2

✓ Visualisation graphique :

```
df_h2.hist(bins=50, figsize=(15, 10))
plt.show()
```



```
df_h1.hist(bins=50, figsize=(15, 10))
plt.show()
```



✓ Harmonisation de l'unité de la colonne Total_Proteins :

```
def check_protein_format(protein_str, unite):
    return isinstance(protein_str, str) and protein_str.endswith(unite)

def extract_protein_value(protein_str):
    if check_protein_format(protein_str, 'mg/dL'):
        protein_float = float(str.split(protein_str, ' ')[0])
        return protein_float
    if check_protein_format(protein_str, 'g/dL'):
        protein_float = float(str.split(protein_str, ' ')[0])
        return protein_float * 1000
    if check_protein_format(protein_str, 'nan mg/dL'):
        return np.nan

df_h1['Total_Proteins'] = df_h1['Total_Proteins'].apply(extract_protein_value)
df_h1.head(3)
```

	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	ALT	AST	Total_Proteins	Albumin	Albumin_and_Globulin_Rat
0	41 to 60 yo	Male	7.0	2.0	174.0	NaN	14	7800.0	4.2	N
1	41 to 60 yo	Male	6.0	2.0	215.0	88.0	24	7100.0	2.4	N

Next steps:

[Generate code with df_h1](#)[View recommended plots](#)[New interactive sheet](#)

✓ Harmonisation de l'unité de Total_Bilirubin, Direct_Bilirubin :

```
def harmonize_Total_Bilirubin(df):
    df['Total_Bilirubin'] = df['Total_Bilirubin']*10
    return df
```

```
df_h2 = harmonize_Total_Bilirubin(df_h2)
df_h2.head(3)
```



	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Total
0	61 to 80 yo	Female	7.0	0.1	187.0		NaN	18
1	61 to 80 yo	Male	109.0	5.5	699.0	64.0		100
2	61 to 80 yo	Male	73.0	4.1	490.0		NaN	68

Next steps:

[Generate code with df_h2](#)

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```
def harmonize_Direct_Bilirubin(df):
    df['Direct_Bilirubin'] = df['Direct_Bilirubin']*10
    return df
```

```
df_h2 = harmonize_Direct_Bilirubin(df_h2)
df_h2.head(3)
```



	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Total
0	61 to 80 yo	Female	7.0	1.0	187.0		NaN	18
1	61 to 80 yo	Male	109.0	55.0	699.0	64.0		100
2	61 to 80 yo	Male	73.0	41.0	490.0		NaN	68

Next steps:

[Generate code with df_h2](#)

 [View recommended plots](#)

[New interactive sheet](#)

✓ Renommer les colonnes du jeu de données df_h1 :

```
df_h1.rename(columns={'Age': 'Age'
    , 'Gender': 'Gender'
    , 'Total_Bilirubin': 'Total_Bilirubin'
    , 'Direct_Bilirubin': 'Direct_Bilirubin'
    , 'Alkaline_Phosphotase': 'Alkaline_Phosphotase'
    , 'ALT': 'Alamine_Aminotransferase'
    , 'AST': 'Aspartate_Aminotransferase'
    , 'Total_Protiens': 'Total_Protiens'
    , 'Albumin': 'Albumin'
    , 'Albumin_and_Globulin_Ratio': 'Albumin_and_Globulin_Ratio'
    , 'Result': 'Outcome'
    }, inplace=True)
```

```
df_h1.head(3)
```



	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Total
0	41 to 60 yo	Male	7.0	2.0	174.0	NaN		14
1	41 to 60 yo	Male	6.0	2.0	245.0	22.0		24
2	21 to 40 yo	Male	11.0	5.0	191.0	37.0		41

Next steps: [Generate code with df_h1](#) [View recommended plots](#) [New interactive sheet](#)

✓ Concaténation des jeux de données :

```
df_combined = pd.concat([df_h1, df_h2])
df_combined
```



	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Tot
0	41 to 60 yo	Male	7.0	2.0	174.0	NaN		14
1	41 to 60 yo	Male	6.0	2.0	245.0	22.0		24
2	21 to 40 yo	Male	11.0	5.0	191.0	37.0		41
3	21 to 40 yo	Male	24.0	10.0	340.0	25.0		21
4	21 to 40 yo	Male	6.0	2.0	202.0	NaN		41
...	...	...	...	...	...	...	...	...
286	21 to 40 yo	Female	8.0	2.0	192.0	15.0		12
287	41 to 60 yo	Male	13.0	6.0	155.0	15.0		20
288	41 to 60 yo	Female	10.0	5.0	239.0	16.0		39
289	41 to 60 yo	Male	45.0	23.0	315.0	120.0		105
290	41 to 60 yo	Female	10.0	3.0	250.0	48.0		44

612 rows × 11 columns

Next steps: [Generate code with df_combined](#) [View recommended plots](#) [New interactive sheet](#)

Retirer les doublons :

```
df_combined.drop_duplicates(inplace=True)
df_combined
```

	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	Alamine_Aminotransferase	Aspartate_Aminotransferase	Tot
0	41 to 60 yo	Male	7.0	2.0	174.0	NaN		14
1	41 to 60 yo	Male	6.0	2.0	245.0	22.0		24
2	21 to 40 yo	Male	11.0	5.0	191.0	37.0		41
3	21 to 40 yo	Male	24.0	10.0	340.0	25.0		21
4	21 to 40 yo	Male	6.0	2.0	202.0	NaN		41
...	...	...	...	...	...	...		...
286	21 to 40 yo	Female	8.0	2.0	192.0	15.0		12
287	41 to 60 yo	Male	13.0	6.0	155.0	15.0		20
288	41 to 60 yo	Female	10.0	5.0	239.0	16.0		39
289	41 to 60 yo	Male	45.0	23.0	315.0	120.0		105
290	41 to 60 yo	Female	10.0	3.0	250.0	48.0		44

606 rows × 11 columns

Next steps:

Generate code with df_combined

View recommended plots

New interactive sheet